

The empirical recurrence for OEIS A254170, recovered from its tail

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8 September 2026

Abstract

OEIS A254170 records “Empirical recurrence of order 73” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 693$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 18$, so the recurrence is proved for every $n \geq 91$. Its last coefficient is $c_{73} = 17179869184 \neq 0$, so no further tail satisfies anything shorter; any order-73 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A254170 is “Number of $(3+2)X(n+2)$ 0..1 arrays with every $3X3$ subblock diagonal maximum minus antidiagonal median nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

16240, 164684, 1553720, 15016048, 139009432, 1263003480, 11377995944, 101942555352, ...

The entry states:

Empirical recurrence of order 73 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:29:44, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 693$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 693$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 73: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 18$ is the first whose minimal order is exactly the stated 73.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1386$ exact terms from $n = 18$ on, it returns order 73 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{73} c_i a(n-i),$$

c_1	12	c_{20}	−2242747312	c_{39}	−4062238720	c_{58}	−50532974592
c_2	−15	c_{21}	1535433600	c_{40}	−32768417792	c_{59}	−33319550976
c_3	−154	c_{22}	1725159344	c_{41}	57822437376	c_{60}	34124857344
c_4	400	c_{23}	−4613578080	c_{42}	23581810688	c_{61}	94422171648
c_5	32	c_{24}	3032986592	c_{43}	−82662211584	c_{62}	79456894976
c_6	−1334	c_{25}	7080415488	c_{44}	−85410054144	c_{63}	−129922760704
c_7	6956	c_{26}	−13244394752	c_{45}	144219176960	c_{64}	−66840428544
c_8	1631	c_{27}	−2410398336	c_{46}	81987174400	c_{65}	129385889792
c_9	−193708	c_{28}	21222937984	c_{47}	−225295859712	c_{66}	−25232932864
c_{10}	420325	c_{29}	−8622112000	c_{48}	82501304320	c_{67}	−105226698752
c_{11}	612406	c_{30}	−27214271232	c_{49}	198569885696	c_{68}	32212254720
c_{12}	−2301862	c_{31}	28731255808	c_{50}	−194229305344	c_{69}	88046829568
c_{13}	1228960	c_{32}	29336703744	c_{51}	−35199123456	c_{70}	8589934592
c_{14}	−6476576	c_{33}	−64248805376	c_{52}	47633661952	c_{71}	−60129542144
c_{15}	91555624	c_{34}	3688791296	c_{53}	−121903251456	c_{72}	−8589934592
c_{16}	−189456792	c_{35}	80950851072	c_{54}	180120190976	c_{73}	17179869184
c_{17}	−332714096	c_{36}	−52247393792	c_{55}	113091018752		
c_{18}	1130868848	c_{37}	−53678129152	c_{56}	−133185929216		
c_{19}	34818240	c_{38}	61591066624	c_{57}	−29536288768		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 91$, and it is the recurrence OEIS A254170 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 18$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{73} - c_1 x^{72} - \dots - c_{73}$. Its constant term is $-c_{73} = -17179869184$, which is not zero, so

$q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 73 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 73, so $\deg q = 73$, and it is monic. It holds from some index t on. From $\max(t, 18)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 73, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 73, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 17179869184.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-73 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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